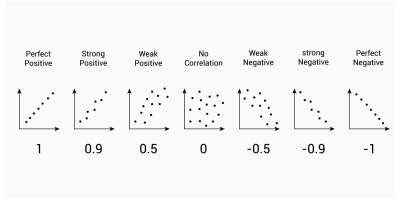


RSR week 06 correlation and regression

1. What does correlation mainly tell us?



a) Direction and strength of a linear relationship ✓

b) The predicted y values for every individual in the data

c) Whether one variable significantly causes the other

2. [SEE IMAGE] In a group of adults, taller people tend to weigh more, as shown. Report the correlation r ?

a) about 1.0

b) about 0.05

c) about 0.8 ✓

d) about -0.5

3. Which of these is NOT a benefit of doing simple linear regression instead of/in a correlation?

a) You get a fitted equation for predicting y from x

b) You get a guarantee that x causes y ✓

c) You get a p value that tells you the probability of results this 'extreme' if there were no relationship

4. In predicted $y = a + b * x$, what does b represent?

$$Y = a + bX + \epsilon$$

a) The slope ✓

b) The outcome variable

c) The beta sample size

5. A simple linear regression has $R\text{-squared} = 0.65$. What is the best interpretation

Variation in outcome (y)

65% explained by x

unexplained

a) The slope equals 0.65 units per 1-unit increase in x

b) About 65% of the variation in y is explained by x in this model ✓

c) 65% of all data points lie exactly on the line

6. In simple linear regression with one predictor, how is $R\text{-squared}$ related to the cc coefficient?

r (correlation)
0.80

$^2 =$

R^2
0.64

a) $R\text{-squared}$ equals the correlation squared (r^2) ✓

b) $R\text{-squared}$ equals the correlation r

c) $R\text{-squared}$ equals 1 minus the correlation

7. [SEE IMAGE] In `summary(model)`, where do you find the fitted intercept and slope

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.00	0.40	5.00	0.001
hours_studied	3.00	0.50	6.00	<0.001

a) The Std. Error column

b) The row labels only (no numbers needed)

c) The Estimate column ✓

